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- Practicalities
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- Course improvements

Bayesian Statistics and Data Analysis
Course information

Måns Magnusson
Department of Statistics, Uppsala University



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Who am I?

1. Associate professor in Statistics
2. Teaching: Machine Learning and Bayesian Statistics
3. Research: Machine Learning and Bayesian Statistics
   - 3.1 Bayesian Statistics and computational statistics



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Who am I?

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   - 3.2 Data-centric machine learning



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   - 3.3 Statistical inference from textual data



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Who am I?

1. Associate professor in Statistics
2. Teaching: Machine Learning and Bayesian Statistics
3. Research: Machine Learning and Bayesian Statistics
   - 3.1 Bayesian Statistics and computational statistics
   - 3.2 Data-centric machine learning
   - 3.3 Statistical inference from textual data
   - 3.4 Applications: Law, sociology, political science, history, and medicine/biology



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Section 1

Course information



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Course information

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Course information

The aims of this course are that, after this course you should:

1. have knowledge in basic concepts, philosophy, and perspectives in Bayesian Statistics,



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Course information

The aims of this course are that, after this course you should:

1. have knowledge in basic concepts, philosophy, and perspectives in Bayesian Statistics,
2. derive posterior distributions in simple situations,



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Course information

The aims of this course are that, after this course you should:

1. have knowledge in basic concepts, philosophy, and perspectives in Bayesian Statistics,
2. derive posterior distributions in simple situations,
3. derive and use predictive distributions,



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The aims of this course are that, after this course you should:

1. have knowledge in basic concepts, philosophy, and perspectives in Bayesian Statistics,
2. derive posterior distributions in simple situations,
3. derive and use predictive distributions,
4. identify and formulate Bayesian probabilistic models for analysis and predictions,



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3. derive and use predictive distributions,
4. identify and formulate Bayesian probabilistic models for analysis and predictions,
5. estimate models using contemporary (state-of-the-art) computer-based methods for posterior approximations,



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4. identify and formulate Bayesian probabilistic models for analysis and predictions,
5. estimate models using contemporary (state-of-the-art) computer-based methods for posterior approximations,
6. understand and use basic principles for decisions under uncertainty,



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5. estimate models using contemporary (state-of-the-art) computer-based methods for posterior approximations,
6. understand and use basic principles for decisions under uncertainty,
7. have knowledge about and be able to use Bayesian methods for model comparisons,



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5. estimate models using contemporary (state-of-the-art) computer-based methods for posterior approximations,
6. understand and use basic principles for decisions under uncertainty,
7. have knowledge about and be able to use Bayesian methods for model comparisons,
8. be able to critically evaluate Bayesian methods,



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7. have knowledge about and be able to use Bayesian methods for model comparisons,
8. be able to critically evaluate Bayesian methods,
9. report, orally and in writing, a Bayesian statistical analysis



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Pre-requisites

- Basic probability theory
  - probability, probability density, distribution
  - sum, product rule, and Bayes' rule
  - expectation, mean, variance, median



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  - expectation, mean, variance, median
- Basic linear algebra and calculus



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Pre-requisites

- Basic probability theory
  - probability, probability density, distribution
  - sum, product rule, and Bayes' rule
  - expectation, mean, variance, median
- Basic linear algebra and calculus
- Basic visualisation techniques (R or Python)
  - histogram, density plot, scatter plot
- *Note!* This is a masters course in Statistics.



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First assignment is a recap.



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Who are you? (very brief)

1. Name
2. What is your expectation on this course?
3. Why did you choose it? (if you chose it)
4. What do you hope to get out of it?



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Course Outline

Main components:

- Core Content (8-9 lecture blocks)



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Course Outline

Main components:

- Core Content (8-9 lecture blocks)
- Assignments (8 individual assignments)



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Course Outline

Main components:

- Core Content (8-9 lecture blocks)
- Assignments (8 individual assignments)
- Oral exams connected to the assignments



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Course Outline

Main components:

- Core Content (8-9 lecture blocks)
- Assignments (8 individual assignments)
- Oral exams connected to the assignments
- Mini-project: own Bayesian data analysis or paper replication track (1-3 students)



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Course Outline

Main components:

- Core Content (8-9 lecture blocks)
- Assignments (8 individual assignments)
- Oral exams connected to the assignments
- Mini-project: own Bayesian data analysis or paper replication track (1-3 students)

Exact dates and details; see the course page.



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Core Content

- Every week: lectures (approx. 2-4h)
  - Online video material and reading assignments (approx. 2-4h, 50-90 pages a week)
  - Lecture(s): present overall theory and content (overview, focus on learning concepts)
  - Assignment(s): Computational and theoretical individual work. (approx. 12-16h). Deadline Sundays 23.59.
    Start monday morning every week!



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  - Lecture(s): present overall theory and content (overview, focus on learning concepts)
  - Assignment(s): Computational and theoretical individual work. (approx. 12-16h). Deadline Sundays 23.59. *Start monday morning every week!*
- Recommended workflow for each week
  - Do the reading assignments
  - Watch the videos (optional)
  - Do self-study exercises
  - Start with the assignment
  - Attend lecture (bring questions!)
  - Attend Zoom datalabs (bring questions! Helps with debugging Stan code)
  - Submit the assignment
  - Review the oral exam questions and practice explaining answers in groups



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Section 2

Assignments



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Assignments

1. Core components and concepts and state-of-the-art methods



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Assignments

1. Core components and concepts and state-of-the-art methods
2. *Warning!* There might be bugs in the assignments!



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Assignments

1. Core components and concepts and state-of-the-art methods
2. *Warning!* There might be bugs in the assignments!
3. All labs can be turned in a three times (See Studium for details), but only first time can give VG points:
   - 3.1 The week of the assignment
   - 3.2 The last day of the course
   - 3.3 2-4 weeks after the course



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4. We will mark and return each assignment hopefully within 5 working days (max 10 working days).



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LLM use and assignment integrity

- New this year! You may use LLMs like fellow students: to learn, ask for explanations, and get feedback.



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- LLMs are not allowed at all for the oral exam questions.



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LLM use and assignment integrity

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- LLMs may not produce submitted solutions.
- You must be able to explain every part of your submitted work orally.
- LLMs are not allowed at all for the oral exam questions.
- Assignment PDFs may contain hidden machine-readable integrity text. Do not copy or upload the full assignment text into any LLM.
- If we suspect that the assignment has been done using an LLM, it will be failed.



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Oral exams

- Three oral exams during the course:
  1. Assignments 1–3
  2. Assignments 4–6
  3. Assignments 7–8



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Oral exams

- Three oral exams during the course:
  1. Assignments 1–3
  2. Assignments 4–6
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- For each covered assignment: one question drawn at random from the oral exam document.



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- Three oral exams during the course:
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- For each covered assignment: one question drawn at random from the oral exam document.
- If the first answer is not sufficient, two new questions are drawn at random from the same assignment. Both must be answered satisfactorily.



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- For each covered assignment: one question drawn at random from the oral exam document.
- If the first answer is not sufficient, two new questions are drawn at random from the same assignment. Both must be answered satisfactorily.
- A failed attempt removes VG eligibility for that assignment, even after a later pass. Missing an opportunity does not.



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- For each covered assignment: one question drawn at random from the oral exam document.
- If the first answer is not sufficient, two new questions are drawn at random from the same assignment. Both must be answered satisfactorily.
- A failed attempt removes VG eligibility for that assignment, even after a later pass. Missing an opportunity does not.
- At least three attempts are available for each oral exam. All assignments' oral exams must be passed by the final opportunity; otherwise, you must retake the course.



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R vs Python and Colab

- We strongly recommend using R in the course as there are more packages for Stan and statistical analysis in general in R



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R vs Python and Colab

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- If you are already fluent in Python, but not in R, then using Python may be easier, but it can still be useful to learn R



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- All data for the assignments are included in the R package. You can access the data `data` or `data-raw` in the R package folder in the repo if you need to get the data without using the `bsda` R package.



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- We supply a Google Colab template with everything pre-installed.



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- We supply a Google Colab template with everything pre-installed.
- We also supply a knitR LaTeX template suitable for Overleaf.

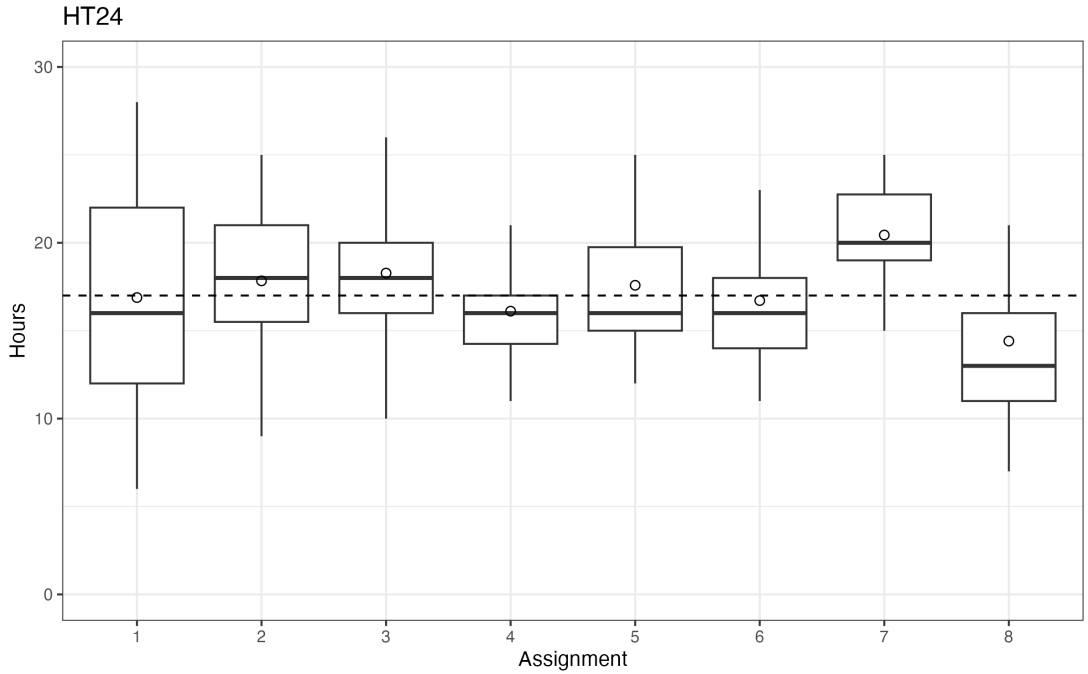


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Course Workload

- Conclusions from 2024
  - Slightly too large workload in assignment 7



Student course workload



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Stan

- Stan is a probabilistic programming framework (PPF) and ecosystem
- 40+ developers, 100+ contributors, 100K+ users
- R, Python, Julia, Scala, Stata, Matlab, command line interfaces
- More than 200 R packages using Stan
- Many packages to support diagnostics and workflow
- Can be used for frequentist inference as well
- Alternative PPF exists, Turing (Julia), Pyro (PyTorch), etc.



mc-stan.org



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Section 3

Mini-project



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Mini-project

- See project instructions on webpage for details.
- Two possible tracks: ordinary data analysis on real data, or paper replication.
- 2-3 students.



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Mini-project

- See project instructions on webpage for details.
- Two possible tracks: ordinary data analysis on real data, or paper replication.
- 2-3 students.
- Supply a one-page project proposal in the middle of the course.



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Mini-project

- See project instructions on webpage for details.
- Two possible tracks: ordinary data analysis on real data, or paper replication.
- 2-3 students.
- Supply a one-page project proposal in the middle of the course.
- Ideally, use a model not presented in this course.
- Project will last two weeks (half time) - but start earlier.
- The project should use Stan.
- Approximate 40 hours of work *per student*.



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Mini-project

- See project instructions on webpage for details.
- Two possible tracks: ordinary data analysis on real data, or paper replication.
- 2-3 students.
- Supply a one-page project proposal in the middle of the course.
- Ideally, use a model not presented in this course.
- Project will last two weeks (half time) - but start earlier.
- The project should use Stan.
- Approximate 40 hours of work *per student*.
- The project should result in a 4 page report (PDF) using the ICML LaTeX template (see course page).
- Project oral presentation (10-15 minutes)



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- The first author is corresponding author



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- The project should result in a 4 page report (PDF) using the ICML LaTeX template (see course page).
- Project oral presentation (10-15 minutes)
- The first author is corresponding author
- *New this year*:
  - Paper replication track: open paper + Stan implementation + reproduce main result + prepare for `posteriordb`.
  - Mini-project discussions: after lectures and during lecture breaks.



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Section 4

Practicalities



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Practicalities

- Course page/content: Github – please do a PR or submit an issue if something is wrong!
  https://github.com/MansMeg/BSDA/issues



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Practicalities

- Course page/content: Github – please do a PR or submit an issue if something is wrong!
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- Communication: Studium



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Practicalities

- Course page/content: Github – please do a PR or submit an issue if something is wrong!
  https://github.com/MansMeg/BSDA/issues
- Communication: Studium
- Schedule: Time Edit/Studium
- Assignments submissions: Studium



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Practicalities

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- Schedule: Time Edit/Studium
- Assignments submissions: Studium
- Acknowledgements: Aki Vehtari



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- Acknowledgements: Aki Vehtari
- Teaching assistant: Väinö Yrjänäinen

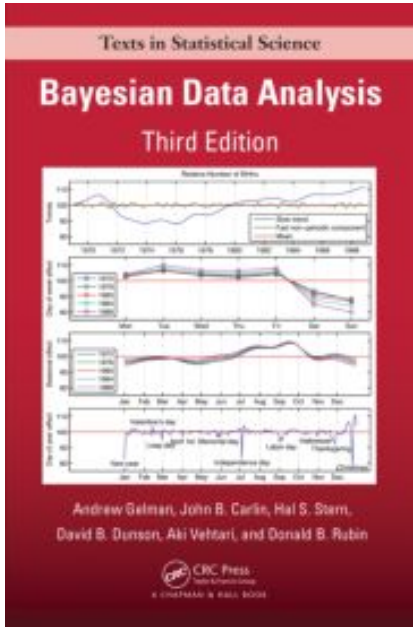


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Literature

- Book: Gelman, Carlin, Stern, Dunson, Vehtari & Rubin: Bayesian Data Analysis, Third Edition. (online pdf available)



- Additional articles, youtube videos and blog posts (see reading list per week)



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Section 5

Examination



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Examination

1. To pass (G): All labs, oral exams, mini-project, and project review need to be passed



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Examination

1. To pass (G): All labs, oral exams, mini-project, and project review need to be passed
2. To pass with distinction (VG): 7/10 VG points



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Examination

1. To pass (G): All labs, oral exams, mini-project, and project review need to be passed
2. To pass with distinction (VG): 7/10 VG points
3. If everything is correct in an assignment (¿90%), 1 VG point is awarded *on the first submission deadline*.



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Examination

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4. Assignment VG points require passing the oral exam for that assignment.



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Examination

1. To pass (G): All labs, oral exams, mini-project, and project review need to be passed
2. To pass with distinction (VG): 7/10 VG points
3. If everything is correct in an assignment (¿90%), 1 VG point is awarded *on the first submission deadline*.
4. Assignment VG points require passing the oral exam for that assignment.
5. The mini-project is worth 2 VG-points (if it is passed with distinction).
6. Ph.D. students: I suggest you get VG to pass the course. Make the project a potential paper.
7. Reassesment of grades (supply form to course admin)
8. Failing the course: You will need to redo all assignments and mini-project next year.
9. LLMs may be used for learning, explanations, and feedback, but not to produce submitted solutions.



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Section 6

Course improvements



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Course improvements since last time

- Oral exams
- New idea (test) of mini project.



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Section 6

Course improvements



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Comments from previous students

- Put in the work, it is worth it.



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Comments from previous students

- Put in the work, it is worth it.
- There is alot to do but it will be worth it if you really put in work.



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Comments from previous students

- Put in the work, it is worth it.
- There is alot to do but it will be worth it if you really put in work.
- Be prepared that the tempo and workload is very high.



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Comments from previous students

- Put in the work, it is worth it.
- There is alot to do but it will be worth it if you really put in work.
- Be prepared that the tempo and workload is very high.
- Attend the lectures and start with the assignments early. Because they are really fun if you get passed the time pressure aspect



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- There is alot to do but it will be worth it if you really put in work.
- Be prepared that the tempo and workload is very high.
- Attend the lectures and start with the assignments early. Because they are really fun if you get passed the time pressure aspect
- Start the assignments as soon as possible. It is perfectly reasonable to be done with the assignments before the weekend, so aim for that.



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Comments from previous students

- Put in the work, it is worth it.
- There is alot to do but it will be worth it if you really put in work.
- Be prepared that the tempo and workload is very high.
- Attend the lectures and start with the assignments early. Because they are really fun if you get passed the time pressure aspect
- Start the assignments as soon as possible. It is perfectly reasonable to be done with the assignments before the weekend, so aim for that.
- Do the course properly, it is worth the effort



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Questions?

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